CAIRNHOLLOW COMMUNITY ENERGY · BOARD PAPER B-26-11-04

Book the 2027 leading-edge campaign on CH04: its blades cost about 89 MWh last year, and the repair should recover 100 MWh to the nearest 50 MWh

To the Board, meeting of 12 November 2026. From Jenna Dyer, Asset Manager, Cairnhollow Community Energy. 14 October 2026. For decision.

Recommendation

Book the summer 2027 rope-access campaign with Skerrow Blade Services on CH04. Measured as OM-07 clause 2.4 asks, against the same turbine in the same winds with its blades as new, CH04's blades cost 89 MWh over its normal operation in the twelve months to September 2026, 1.3 per cent of what it produced. The repair should recover about that over the twelve months after the work: 100 MWh to the nearest 50 MWh. The next largest loss at the same comparison is CH02's, 65 MWh, which makes CH02 the candidate for 2028.

Why not the turbine with the largest shortfall

Against the warranted power curve at the same mean wind, CH07 falls 235 MWh short, more than twice any other turbine, and CH01 and CH05 follow. None of that is blade loss. A ten-minute power is the power curve averaged over the gusts inside the ten minutes; the three turbines facing the plantation run in its eddies (CH07 at 2.3 times the turbulence of the turbines in open air, at 8 to 12 m/s in south-westerly and westerly winds), so near rated power the gusts swing them below the curve's knee and they make less, while between 4 and 8.5 m/s the same gusts lift them above the warranted curve, which no blade does. Compared with an as-new turbine at each ten-minute record's own turbulence (the turbulence normalisation of IEC 61400-12-1 Annex M, from each nacelle anemometer's standard deviation), CH07, CH05 and CH01 lose less than 1 MWh each: their blades are sound, and a repair would recover nothing.

Why not CH02, and why CH04's loss was hidden

Below rated, where erosion is usually looked for, CH02 shows the largest deficit (77 MWh between 3 and 10 m/s against 42 for CH04). Part of it is real: CH02's blades lose 65 MWh at the same comparison. The rest is the loch: CH02 has the smoothest air on the site in westerly winds, and in smooth air a ten-minute power at low wind sits below a curve measured in ordinary turbulence. CH04 stands three and a half rotor diameters downwind of CH02 in westerly winds, and the wake's turbulence lifts its low-wind power by about as much as its blades lose there (its shortfall between 4 and 8 m/s is 0.5 MWh); its loss shows near rated and at its own turbulence.

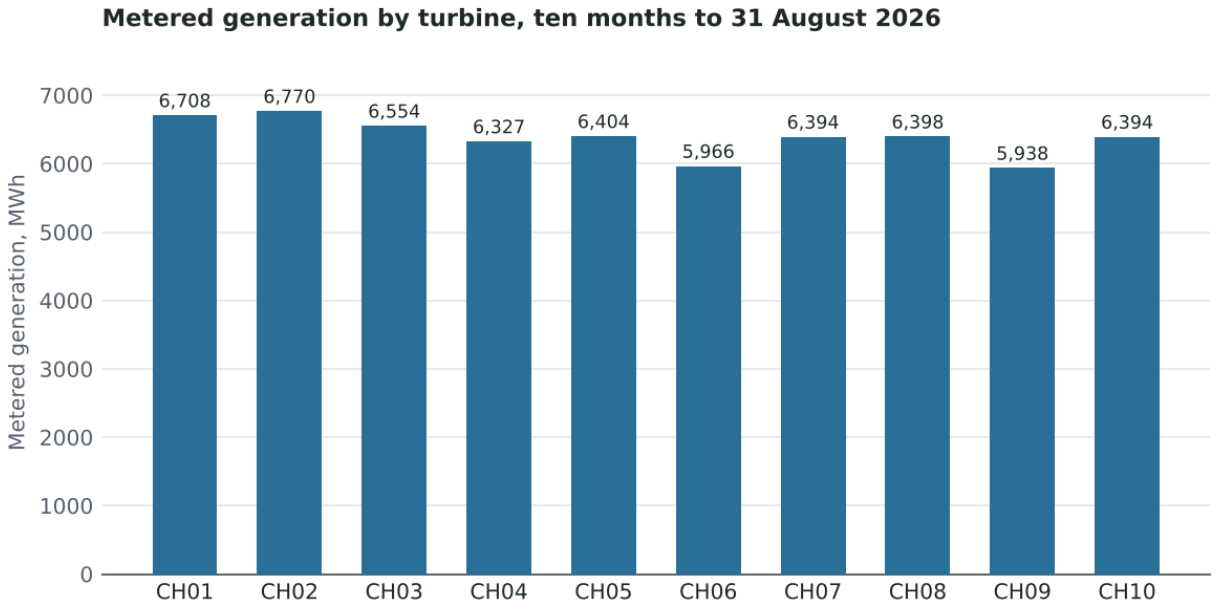
CH09's blades cost 361 MWh over the whole year, but those were the blades of the rotor lowered for the main-bearing exchange in April; under OM-07 clause 2.5 it is measured on the pool rotor in service since 25 April 2026, whose blades are sound.

Turbine	Shortfall against the warranted curve at the same mean wind (MWh)	Blade loss at the same turbulence (MWh)	Metered generation, ten months to 31 August (MWh)
CH01	115	0	6,708
CH02	71	65	6,770
CH03	3	0	6,554
CH04	93	89	6,327
CH05	114	0	6,404
CH06	4	0	5,966
CH07	235	1	6,394
CH08	2	0	6,398
CH09	-3	0	5,938
CH10	9	0	6,394
Farm			63,853

Normal operation (states 10 and 20), October 2025 to September 2026; CH09 from 25 April 2026. Sources: SCADA, met mast M1, the monthly round's counter reads, work orders, Schedule 3.

What the meters recorded, 2025-26

The ten turbines generated 63,853 MWh over the ten months to 31 August 2026 as their own revenue counters record it, and the connection meter exported 60,591 MWh over the same months. The counters are read on the monthly round rather than at the month end, so each turbine's figure is taken to the window's own ends from the ten-minute record. The workbook metered_generation_2025_26.xlsx gives each turbine's generation and the export month by month.



Notes

1. Blade loss is the as-new power at each normal-operation record's own wind and turbulence less the power made, summed over the twelve months. The as-new curve is the zero-turbulence curve that reproduces Schedule 3 at the mast's turbulence (power coefficient 0.470, cut-in 3.0 m/s, rated 2,300 kW), averaged over each record's turbulence. Comparing each turbine with the fleet in the same 0.5 m/s and turbulence bins gives 83 MWh for CH04.

2. Wind speeds are corrected to 1.225 kg/m3 with the mast's temperature and pressure. The SCADA export joins the site server's archive, stamped in UK local time, to the hosted platform's export, stamped in UTC from 3 February 2026; it is aligned to the mast and the ANM log on that basis.

3. Downtime includes the hours a turbine's controller was isolated for its annual service or a repair, which the SCADA does not record; the work orders give them. The site communications outage of 18 and 19 November 2025 (turbines running, counters recording) and the grid loss of 24 January 2026 are neither downtime nor curtailment.